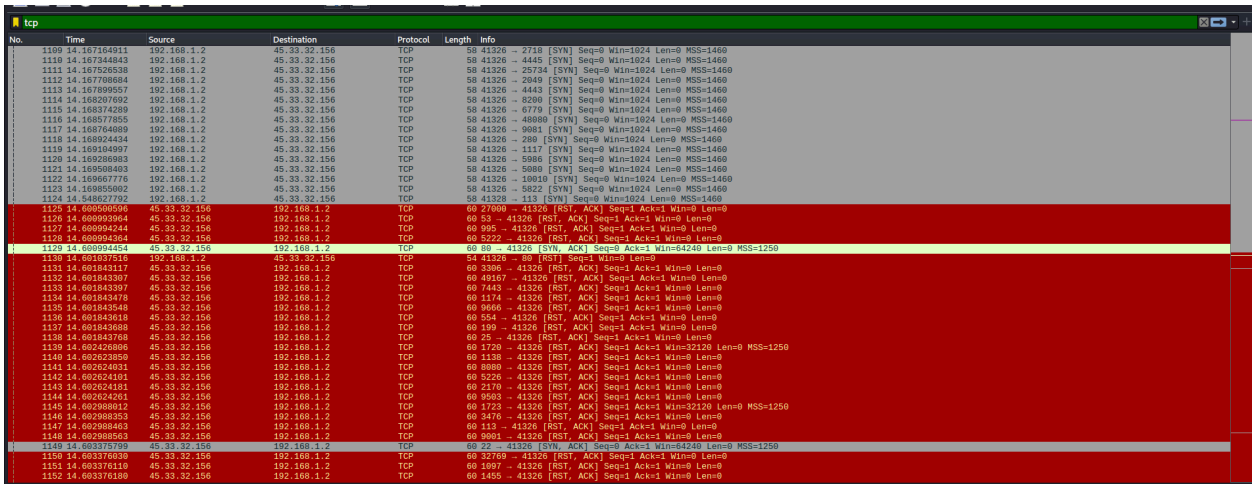
Attack Simulation + Traffic Detection

Port Scanning using nmap:

I did a nmap scan on domain: scanme.nmap.org - commands use: ~ nmap scanme.nmap.org

- What i found:
  - Too many SYN packets with no SYN,ACK responses
  - Too many packets with RST flag set
  - SYN packets are on different port numbers
  - Incomplete Handshake



tcp

No.	Time	Source	Destination	Protocol	Length	Info
1109	14.167164911	192.168.1.2	45.33.32.156	TCP	58	41326 → 2718 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1110	14.167344843	192.168.1.2	45.33.32.156	TCP	58	41326 → 4445 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1111	14.167526538	192.168.1.2	45.33.32.156	TCP	58	41326 → 25734 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1112	14.167708684	192.168.1.2	45.33.32.156	TCP	58	41326 → 2049 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1113	14.167899557	192.168.1.2	45.33.32.156	TCP	58	41326 → 4443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1114	14.168207692	192.168.1.2	45.33.32.156	TCP	58	41326 → 8200 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1115	14.168374289	192.168.1.2	45.33.32.156	TCP	58	41326 → 6779 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1116	14.168577855	192.168.1.2	45.33.32.156	TCP	58	41326 → 48080 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1117	14.168764089	192.168.1.2	45.33.32.156	TCP	58	41326 → 9081 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1118	14.168924434	192.168.1.2	45.33.32.156	TCP	58	41326 → 280 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1119	14.169104997	192.168.1.2	45.33.32.156	TCP	58	41326 → 1117 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1120	14.169286983	192.168.1.2	45.33.32.156	TCP	58	41326 → 5986 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1121	14.169508403	192.168.1.2	45.33.32.156	TCP	58	41326 → 5080 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1122	14.169667776	192.168.1.2	45.33.32.156	TCP	58	41326 → 10010 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1123	14.169855002	192.168.1.2	45.33.32.156	TCP	58	41326 → 5822 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1124	14.548627792	192.168.1.2	45.33.32.156	TCP	58	41326 → 113 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1125	14.600500596	45.33.32.156	192.168.1.2	TCP	60	27000 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1126	14.600993964	45.33.32.156	192.168.1.2	TCP	60	53 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1127	14.600994244	45.33.32.156	192.168.1.2	TCP	60	995 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1128	14.600994364	45.33.32.156	192.168.1.2	TCP	60	5232 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1129	14.600994454	45.33.32.156	192.168.1.2	TCP	60	80 → 41326 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1250
1130	14.601037516	192.168.1.2	45.33.32.156	TCP	54	41326 → 80 [RST] Seq=1 Win=0 Len=0
1131	14.601843117	45.33.32.156	192.168.1.2	TCP	60	3306 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1132	14.601843307	45.33.32.156	192.168.1.2	TCP	60	49167 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1133	14.601843397	45.33.32.156	192.168.1.2	TCP	60	7443 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1134	14.601843478	45.33.32.156	192.168.1.2	TCP	60	1174 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1135	14.601843548	45.33.32.156	192.168.1.2	TCP	60	9666 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1136	14.601843618	45.33.32.156	192.168.1.2	TCP	60	554 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1137	14.601843688	45.33.32.156	192.168.1.2	TCP	60	199 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1138	14.601843768	45.33.32.156	192.168.1.2	TCP	60	25 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1139	14.602426806	45.33.32.156	192.168.1.2	TCP	60	1720 → 41326 [RST, ACK] Seq=1 Ack=1 Win=32120 Len=0 MSS=1250
1140	14.602623850	45.33.32.156	192.168.1.2	TCP	60	1138 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1141	14.602624031	45.33.32.156	192.168.1.2	TCP	60	8080 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1142	14.602624101	45.33.32.156	192.168.1.2	TCP	60	5226 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1143	14.602624181	45.33.32.156	192.168.1.2	TCP	60	2170 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1144	14.602624261	45.33.32.156	192.168.1.2	TCP	60	9503 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1145	14.602908012	45.33.32.156	192.168.1.2	TCP	60	1723 → 41326 [RST, ACK] Seq=1 Ack=1 Win=32120 Len=0 MSS=1250
1146	14.602988353	45.33.32.156	192.168.1.2	TCP	60	3476 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1147	14.602988463	45.33.32.156	192.168.1.2	TCP	60	113 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1148	14.602988563	45.33.32.156	192.168.1.2	TCP	60	9001 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1149	14.603376139	45.33.32.156	192.168.1.2	TCP	60	22 → 41326 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1250
1150	14.603376030	45.33.32.156	192.168.1.2	TCP	60	32769 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1151	14.603376110	45.33.32.156	192.168.1.2	TCP	60	1097 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1152	14.603376180	45.33.32.156	192.168.1.2	TCP	60	1455 → 41326 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

Analysis:

I found it suspicious because there's too many SYN packets without SYN,ACK in return, and SYN packets are on too many different port numbers which indicates port scanning, theres incomplete handshake too.

If SYN packets are responded with SYN,ACK packets meaning its open and RST packets are responded then it means its closed

DNS Suspicious Traffic:

I run a nslookup command to a very long random domain name simulating a data tunneling behaviour or data exfiltration, command: ~ nslookup verylong…….com

- What i found in wireshark:
  - Multiple packets with long domain name queries
  - Domain name containing random string in it
  - Same domain is queried multiple times

Length	Info
107	Standard query 0x5b34 A verylongrandomstringexampletestdomain123456.com
180	Standard query response 0x5b34 No such name A verylongrandomstringexampletestdomain123456.com SOA a.gtld-servers.net
107	Standard query 0xb02b A verylongrandomstringexampletestdomain123456.com
180	Standard query response 0xb02b No such name A verylongrandomstringexampletestdomain123456.com SOA a.gtld-servers.net
107	Standard query 0x5e7b A verylongrandomstringexampletestdomain123456.com
180	Standard query response 0x5e7b No such name A verylongrandomstringexampletestdomain123456.com SOA a.gtld-servers.net
107	Standard query 0xadfe A verylongrandomstringexampletestdomain123456.com
180	Standard query response 0xadfe No such name A verylongrandomstringexampletestdomain123456.com SOA a.gtld-servers.net
107	Standard query 0xaf04 A verylongrandomstringexampletestdomain123456.com
107	Standard query 0xd678 A verylongrandomstringexampletestdomain123456.com
180	Standard query response 0xd678 No such name A verylongrandomstringexampletestdomain123456.com SOA a.gtld-servers.net
180	Standard query response 0xaf04 No such name A verylongrandomstringexampletestdomain123456.com SOA a.gtld-servers.net
208	Destination unreachable (Port unreachable)
107	Standard query 0xb08e A verylongrandomstringexampletestdomain123456.com
107	Standard query 0xb08e A verylongrandomstringexampletestdomain123456.com
107	Standard query 0xb08e A verylongrandomstringexampletestdomain123456.com
180	Standard query response 0xb08e No such name A verylongrandomstringexampletestdomain123456.com SOA a.gtld-servers.net

Analysis:

Suspicious activity is going on as there is a repeated query to a particular domain name which is a very long random string characters, possible a sign of data exfiltration and data tunneling

Traffic Burst/ Dos Pattern:

I pinged my main Host machine from VM to simulate a Dos Pattern, command used:
~ sudo hping3 -S --flood --rand-source -p 80 192.168.1.143

- What i found:
  - Too many TCP packet in a fraction of seconds
  - All having only SYN packets
  - 17267 number of packets in a fraction of second
  - Destination is same
  - Incomplete Handshake

No.	Time	Source	Destination	Protocol	Length	Info
7037	2.382355653	217.240.132.97	192.168.1.143	TCP	54	9242 → 80 [SYN] Seq=0 Win=512 Len=0
7038	2.382506890	78.79.109.57	192.168.1.143	TCP	54	9243 → 80 [SYN] Seq=0 Win=512 Len=0
7039	2.382681632	186.47.44.105	192.168.1.143	TCP	54	9244 → 80 [SYN] Seq=0 Win=512 Len=0
7040	2.382853378	29.194.54.13	192.168.1.143	TCP	54	9245 → 80 [SYN] Seq=0 Win=512 Len=0
7041	2.383005628	8.132.161.222	192.168.1.143	TCP	54	9246 → 80 [SYN] Seq=0 Win=512 Len=0
7042	2.383157166	33.86.77.194	192.168.1.143	TCP	54	9247 → 80 [SYN] Seq=0 Win=512 Len=0
7043	2.383309184	203.130.109.113	192.168.1.143	TCP	54	9248 → 80 [SYN] Seq=0 Win=512 Len=0
7044	2.383460883	242.36.222.88	192.168.1.143	TCP	54	9249 → 80 [SYN] Seq=0 Win=512 Len=0
7045	2.383604125	141.130.171.98	192.168.1.143	TCP	54	9250 → 80 [SYN] Seq=0 Win=512 Len=0
7046	2.383780209	239.34.14.117	192.168.1.143	TCP	54	9251 → 80 [SYN] Seq=0 Win=512 Len=0
7047	2.383972505	77.167.240.60	192.168.1.143	TCP	54	9252 → 80 [SYN] Seq=0 Win=512 Len=0
7048	2.384274569	67.176.8.14	192.168.1.143	TCP	54	9253 → 80 [SYN] Seq=0 Win=512 Len=0
7049	2.384431517	60.3.66.168	192.168.1.143	TCP	54	9254 → 80 [SYN] Seq=0 Win=512 Len=0
7050	2.384616889	3.53.221.242	192.168.1.143	TCP	54	9255 → 80 [SYN] Seq=0 Win=512 Len=0
7051	2.384798444	93.239.124.86	192.168.1.143	TCP	54	9256 → 80 [SYN] Seq=0 Win=512 Len=0
7052	2.384952366	117.197.47.137	192.168.1.143	TCP	54	9257 → 80 [SYN] Seq=0 Win=512 Len=0
7053	2.385106600	196.239.194.13	192.168.1.143	TCP	54	9258 → 80 [SYN] Seq=0 Win=512 Len=0
7054	2.385264099	237.222.193.100	192.168.1.143	TCP	54	9259 → 80 [SYN] Seq=0 Win=512 Len=0
7055	2.385410948	210.124.211.146	192.168.1.143	TCP	54	9260 → 80 [SYN] Seq=0 Win=512 Len=0
7056	2.385601921	27.119.47.14	192.168.1.143	TCP	54	9261 → 80 [SYN] Seq=0 Win=512 Len=0
7057	2.385769740	91.37.237.125	192.168.1.143	TCP	54	9262 → 80 [SYN] Seq=0 Win=512 Len=0
7058	2.385937488	113.86.29.69	192.168.1.143	TCP	54	9263 → 80 [SYN] Seq=0 Win=512 Len=0
7059	2.386089988	88.54.132.192	192.168.1.143	TCP	54	9264 → 80 [SYN] Seq=0 Win=512 Len=0
7060	2.386242668	31.20.52.6	192.168.1.143	TCP	54	9265 → 80 [SYN] Seq=0 Win=512 Len=0
7061	2.386395559	70.13.60.151	192.168.1.143	TCP	54	9266 → 80 [SYN] Seq=0 Win=512 Len=0
7062	2.386548219	204.210.218.6	192.168.1.143	TCP	54	9267 → 80 [SYN] Seq=0 Win=512 Len=0
7063	2.386701710	212.0.19.136	192.168.1.143	TCP	54	9268 → 80 [SYN] Seq=0 Win=512 Len=0
7064	2.386855432	86.197.109.86	192.168.1.143	TCP	54	9269 → 80 [SYN] Seq=0 Win=512 Len=0
7065	2.387008163	255.30.86.6	192.168.1.143	TCP	54	9270 → 80 [SYN] Seq=0 Win=512 Len=0
7066	2.387219644	230.69.117.50	192.168.1.143	TCP	54	9271 → 80 [SYN] Seq=0 Win=512 Len=0
7067	2.387368728	203.167.74.29	192.168.1.143	TCP	54	9272 → 80 [SYN] Seq=0 Win=512 Len=0

Analysis:

Too many TCP packet around 17K in a fraction of seconds on a same destination resembles Dos attack on a machine/server, they are also only SYN packets without SYN,ACk packets which also resembles attack/suspicious behaviour - (incomplete handshake)